

Neural Edge Full Application Tickets (PDF Version)

Phase 1: Core Infrastructure & Services

Ticket 1: Project & Folder Structure - Type: Task - Priority: High - Description: Create base folders for full app: origin-server, edge-server, edge-functions, sdk, admin-panel, analytics. Initialize package.json and Dockerfile placeholders. - Acceptance Criteria: All folders exist with placeholders and Dockerfiles. - Assignee: DevOps

Ticket 2: Origin Server – Full Content API - Type: Task - Priority: High - Description: Serve static content, assets, and full API endpoints. Implement authentication and rate limiting. - Acceptance Criteria: Origin server serves static + dynamic content with authentication support. - Assignee: Backend

Ticket 3: Edge Server – Caching & Prediction - Type: Task - Priority: High - Description: Serve static content from cache. Fetch missing content from origin. Log cache hits/misses and request latency. Implement AI-based predictive caching. - Acceptance Criteria: Edge server serves cached content, logs events, supports predicted preloading. - Assignee: Backend

Ticket 4: Dynamic Edge Functions - Type: Task - Priority: Medium - Description: Implement API endpoints at edge: /api/user-profile, /api/dashboard, etc. Include logging and optional caching. - Acceptance Criteria: Dynamic APIs served from edge with correct JSON and cache headers. - Assignee: Backend

Ticket 5: SDK – Frontend Integration - Type: Task - Priority: Medium - Description: SDK functions getStatic(), getDynamic(), auto-cache management. Integrate with JS, React, Vue, or Next.js frontend. - Acceptance Criteria: SDK can fetch static/dynamic content, handle edge caching seamlessly. - Assignee: Frontend

Phase 2: DevOps & Containerization

Ticket 6: Dockerize Full Application - Type: Task - Priority: Medium - Description: Create docker-compose.yml to run all services (origin, edge, edge-functions, analytics, admin). Include network and volume configuration. - Acceptance Criteria: docker-compose up starts all services successfully. - Assignee: DevOps

Ticket 7: CI/CD Setup - Type: Task - Priority: Medium - Description: GitHub Actions / GitLab CI to build, test, and deploy containers. Include automated test of SDK and edge caching. - Acceptance Criteria: Code pushed → containers build, tests pass, deployment ready. - Assignee: DevOps

Phase 3: Testing & Analytics

Ticket 8: Local & Load Testing - Type: Task - Priority: High - Description: Automated script to hit static + dynamic endpoints multiple times. Log cache hits/misses and measure latency. Simulate multiple users. -

Acceptance Criteria: Generate performance report comparing origin-only vs edge delivery. - Assignee: QA / Dev

Ticket 9: Monitoring & Logging - Type: Task - Priority: Medium - Description: Add centralized logging for edge and origin servers. Implement basic analytics dashboard (request count, cache hit rate, latency). - Acceptance Criteria: Logs accessible; dashboard shows real-time stats. - Assignee: DevOps / Backend

Phase 4: Admin & Advanced Features

Ticket 10: Admin Panel – Cache Management - Type: Task - Priority: Medium - Description: Admin UI to view cache, purge files, set TTLs. - Acceptance Criteria: Admin can see cached files and perform cache operations. - Assignee: Frontend / Backend

Ticket 11: AI-Powered Predictive Caching (v2) - Type: Feature - Priority: Low - Description: Use AI to predict most-requested content per region. Preload predicted content to edge nodes. - Acceptance Criteria: System pre-caches predicted content, measurable hit rate improvement. - Assignee: Backend / Data Science

Ticket 12: SDK Enhancements - Type: Feature - Priority: Medium - Description: Add offline support, retry logic, analytics reporting from client. - Acceptance Criteria: SDK robust for production integration. - Assignee: Frontend

Ticket 13: Documentation & Tutorials - Type: Task - Priority: Low - Description: Full README, setup guides, SDK docs, and demo instructions. - Acceptance Criteria: Clear instructions to run MVP and full application locally or in cloud. - Assignee: Tech Writer / Dev

Ticket 14: Quantum-Inspired Predictive Caching - Type: Feature - Priority: Medium → High - Description: Implement a quantum-inspired optimization module to enhance predictive caching and dynamic request routing. Use QAOA-inspired heuristics or simulated annealing to decide which content to pre-cache at each edge node. - Acceptance Criteria: Module runs in background or periodically. Edge servers preload content according to results. Logs show improved cache hit rates and reduced latency. - Assignee: Backend / AI Engineer - Phase: Advanced Features / Phase 4–5

Phase 5: Optional Enhancements

- Geo-based intelligent routing
- Real-time content invalidation
- Analytics alerts (cache miss thresholds, latency spikes)
- Multi-cloud edge deployment